Vision-Based Predecessor-Following Robot Convoys for Flexible Material Transport

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Code and evidence

Objective

Can one route-aware robot guide independent cargo carts through warehouse aisles while each follower sees only the robot ahead?

Test task: receiving dock to dispatch dock through mapped shelf aisles.

Approach

LEADER PLANS THE ROUTE

LiDAR, AMCL, Nav2 Smac Hybrid-A*, and Regulated Pure Pursuit.

FOLLOWERS TRACK THE ROBOT AHEAD

Each cart uses forward RGB and local wheel odometry. Followers receive no map, LiDAR, planned path, or Gazebo truth.

WHEN THE TARGET DISAPPEARS

TRACK, PREDICT, SAFE STOP, bounded SEARCH, then confirmed REACQUIRE. Search has zero forward command and a 0.20 rad/s yaw limit.

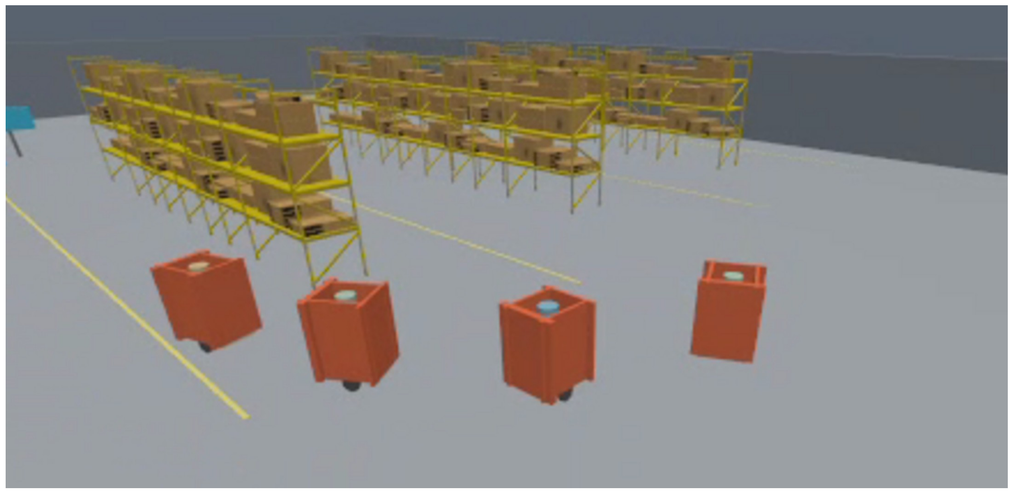
SEALED STATIC CAMERA GATE

YOLOv8n v3: 100% valid full-visible samples; 3.23 cm range RMSE; 0.157° bearing MAE; 9.02 ms inference p95.

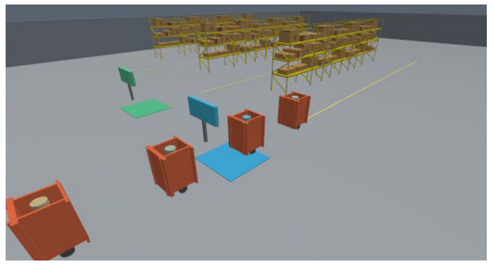
Results

10/10 routes completed

One controlled blackout plus nine mapped missions. Zero sampled collisions; four mapped runs passed every strict check.



Mapped aisle turn in Gazebo. This frame illustrates a presentation run.



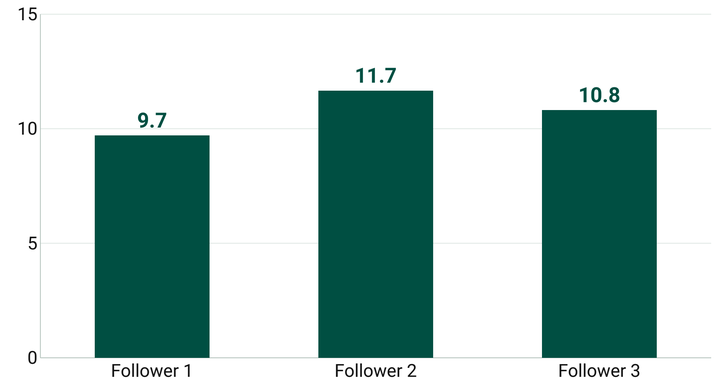
Receiving: scripted cargo cue



Dispatch: simulated unload cue

Nominal convoy spacing error

Three repeated full missions, mean RMSE (cm)



Strict evaluator checks

Mapped group	Strict pass	Min TRACK
Pair, n=3	3/3	95.24%
Chain, n=3	1/3	89.01%
Dropout, n=3	0/3	83.15%

Five chain runs missed only the unchanged 90% TRACK-time check.

Conclusions

Demonstrated

One route-owning leader guided three independent carts through controlled Gazebo warehouse routes. Spacing limits held, while recovery reduced TRACK time in five chain runs.

Future work and application

Improve target visibility, then test physical carts with changing light and moving obstacles. Small or midsize airport baggage and inspection-cart transport is a future application hypothesis.

Simulation only: no physical robot, person-safety, airport, robotic loading, throughput, or savings evidence.

References

[1] Kanchani, EE 616 project repository and November 9 evidence, 2026. [2] Nav2, Smac Planner and Regulated Pure Pursuit documentation.

github.com/justneeraj12/ee616-vision-leader-follower